KEMPE CHAINS AND HADWIGER'S CONJECTURE

REPORT MADE BY ALEX ZHENG AFTER W2026 URA

Abstract. This is a report made after the opportunity to join and learn from Professor Nehaniv's group at the University of Waterloo in Winter (Jan. - Apr.) 2026. [1] [3].

- Reiterate information on Kempe Chains (Kempe's 4CT Method), graph minors, graph degeneracy, Mader's Theorems
- Introduce a fun proposition that a modified Kempe Chain algorithm can demonstrate Hadwiger's Conjecture. This is just a fun proposition, which is highly likely false.
- Propose that for $2 \leq t \leq 3$, $4 \leq t \leq 7$, graphs with no K_t minor (which are k-degenerate) have at least $(k+1)$, $(t-1)$ vertices with degree $\geq k$, respectively.
- Show the counterexamples that show why nonplanar graphs fail with the original Kempe chain approach but succeed for the modified Kempe Chain approach.
- (Think Hadwiger's Conjecture 2.1 is **maybe** true now)

Important: Since I'm making this document public: Thanks to Prof. Nehaniv, Amena, Hanna, who taught me and gave me seminars on math – graphs and algebra. They gave me the info necessary to make this report, that is why this is a report for them and I don't officially cite them. Also, if I made some plagiarism, please let me know; since this is not official, so therefore unofficial, I can make a change.

The code and videos for the modified Kempe Chain approach can be interesting to see. Always a work in progress. Repository link.

If anybody interested would like a quick 10 min explanation of this URA report. a58zheng@uwaterloo.ca.

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1. Describing the Original Flawed Kempe Method (1879)

The Four-Color Theorem (4CT) says that any planar, simple graph can be colored by at most 4 colors. Kempe created this false 'proof' to the 4CT in 1879. It was disproved in 1890 by Heawood with a counterexample. We will describe Kempe's original (inductive) method and show the counterexample.

The original Kempe method for simple planar graphs works on the basis of the principle of 5-degeneracy of planar graphs. The 5-degeneracy of planar graphs can be derived in multiple ways, including Euler's Formula and graph minors. In the first section, we show this using Euler's Formula and later using graph minors.

1.1. **5-degeneracy of planar graphs using Euler's Formula.**

Theorem 1.1. *(Euler's Formula)*
For non-null, ***connected*** *planar graphs,*

$$r = e - v + 2$$

Where r is the number of regions, v is the number of vertices, and e is the number of edges.

Proof. (Euler's Formula)
Intuitive and very standard, but so powerful. □

Definition 1.2. *(Triangulated/Maximal planar graphs)*
From [8]*, a simple planar graph G is said to be* ***triangulated*** *(also called maximal planar) if the addition of any edge to G results in a nonplanar graph. Every finite region is bounded by 3 edges, since the graph is* ***simple****.*

Definition 1.3. *(Graph Degeneracy)*
From [7]*, the* ***degeneracy*** *of a graph G is defined as the smallest integer k such that each subgraph of G contains a vertex of degree at most k.*

Lemma 1.4. *(Handshaking Lemma)*
For any undirected graph $G = (V, E)$, the following identity holds:

$$\sum_{v \in V} \deg(v) = 2e$$

Proof. (Handshaking Lemma)
The idea is that the sum of degrees of all vertices counts each edge twice. □

Theorem 1.5. *(Planar graphs are 5-degenerate through Euler's Formula 1.1)*
Let G be a non-null simple connected planar graph. The average degree, $\mathrm{AvgDeg}(G)$*, is less than 6, or G is 5-degenerate.*

Proof. (Planar graphs are 5-degenerate through Euler's Formula)
It is trivial to show that $\mathrm{AvgDeg}(G) \leq 5$ for $v = 1, 2$.

Let G be a connected **triangulated** graph with $v \geq 3$. Each finite region is bounded by 3 edges and the infinite region has ≥ 3 edges. Every edge is shared between two regions. Hence,

$$2e \geq 3r$$

The number of edges of each region is 3 or more (due to infinite region). So when we count the number of edges via regions, counting every edge twice, we find that $3r$ is equal to or an underestimate of $2e$.

Note: Similarly to the Handshaking Lemma 1.4, each edge is counted twice.

Then, using Euler's Formula 1.1 since the graph is non-null and connected, we get

$$2e \geq 3(e - v + 2)$$

And simplifying yields

$$e \leq 3v - 6$$

Since planar graphs are subgraphs of a triangulation with 0 or more edges removed, we get an edge bound for all planar graphs:

$$\boxed{e \leq 3v - 6}$$

The average degree of the vertices of any graph is given by:

$$\text{AvgDeg}(G) = \frac{\sum_{v \in V} \deg(v)}{v}$$

By the Handshaking Lemma 1.4, we obtain:

$$\text{AvgDeg}(G) = \frac{2e}{v}$$

So for all connected simple planar graphs the AvgDeg is:

$$\text{AvgDeg}(G) \leq \frac{2(3v-6)}{v}$$
$$\text{AvgDeg}(G) \leq 6 - \frac{12}{v}$$

Since $v > 0$ for our non null graph $(V(G) \in \mathbb{W})$, in fact $v \geq 3$. And the $\text{AvgDeg}(G) \geq 0 \forall G$

$$\boxed{0 \leq \text{AvgDeg}(G) \leq 5}$$

So, AvgDeg of any simple non-null connected planar graph G is $\text{AvgDeg}(G) \leq 5$. Hence, any non-null simple planar graph is 5-degenerate.

□

1.2. **Kempe's Method (4CT).**

We now have the information we need to explain Kempe's original flawed inductive 4CT proof.

Let G be a connected non-null simple planar graph $\implies$ $\text{AvgDeg}(G) \leq 5$, from 1.5. This means $\exists$ a vertex in every non-null simple planar graph with $(0 \leq)$degree ≤ 5.

Kempe's inductive method decomposes G inductively by identifying this vertex with degree ≤ 5 and applying a case-by-case method for all 5 degree cases. This method is applied recursively until G is colored.

We will first discuss the 5-case Kempe method and then the Proof by Induction that follows.

1.3. **5-Case Kempe Chain Method.**

Let G be a connected non-null simple planar graph. Suppose $\exists$ an uncolored vertex v_* in G with degree ≤ 5. Let all vertices $\{v \in V(G) \setminus v_*\}$ be properly 4-coloured. So all the neighbours of v_* are colored.

This case-by-case Kempe method shows that v_* can be coloured so that G is 4-colored.

Proof. (Kempe Chains... excluding Heawood Counterexample)

$$\text{Let's define the set: } Colours = \{Red, Green, Blue, Yellow\}.$$

$$\text{Let's define. } \mathrm{Col}(v) : \text{returns the color of } v$$

$$\text{Let's define. } \delta(v) : \text{returns the degree of } v$$

Definition 1.6. *(Kempe Chain)*

A Kempe Chain ***of*** *a vertex v in G is the maximal connected (induced) subgraph of (two) colors, Col(v) and another color, including the vertex v. The only colors are Col(v) and another color. For example, if we take a vertex v colored B and take the B-Y the B-Y Kempe chain on v, it consists of all vertices coloured B or Y that can be reached from v along paths containing only B or Y nodes.*

"Any edge coming out of this subgraph can only connect to a vertex not colored B or Y (by its maximality) so the adjacent vertices to the Kempe Chain subgraph will necessarily be colored R or G." From [5].

This definition was actually created by Professor.[5].

Definition 1.7. *(Kempe Chain Toggling)*

Toggling a Kempe Chain means swapping the two colors of a Kempe Chain. Given a Kempe Chain G' (induced connected subgraph of G) colored only C_1 and C_2, toggling G' colors the vertices in G' the opposite color. So, C_1-colored vertices become C_2, and vice versa. A Kempe Chain is usually said to be toggled at one of its vertices $v \in G'$.

Super important: The intuition behind Kempe chains is as follows. Suppose that two neighbouring vertices of v are differently colored. If there is NO Kempe chain between these two vertices, we can toggle the Kempe chain at one of these neighbouring vertices and free up a color in the graph. Now it becomes obvious why we do this: to be more frugal with our colors.

(Cases)

1. **Case 1:** $\underline{\delta(v_*) \leqslant 3}$ **or** $\{\mathbf{Col}(\mathbf{Neighbours}(v_*))\} \neq Colours$**.**
   In these cases, the uncolored vertex $v_* \in V(G)$ can be colored any color $\{c \in Colours \mid c \notin \{\mathrm{Col}(\mathrm{Neighbours}(v_*))\}\}$.
2. **Case 2:** $\delta(v_*) = 4$ **and** $\{\mathbf{Col}(\mathbf{Neighbours}(v_*))\} = Colours$**.**
   Start with any of the colored neighbours of v_* and define it v_{start}.
   (a) If the Kempe Chain with colors Col(v_{start}) and C_{choose} which includes v_{start} does not include the neighbor of v_*, that is colored C_{choose}, then toggle the Col(v_{start})—C_{choose} Kempe Chain at v_{start}.
   Since v_{start} is now colored C_{choose}, color v_* Col$_{original}(v_{start})$, i.e., the color of the neighboring node whose Kempe Chain was just *toggled*.
   (b) If the Kempe Chain with colors Col(v_{start}) and C_{choose} did include the neighbor of v_* colored C_{choose}, then toggling the Kempe chain would be useless. The neighbours of v_* would still be the complete set: $Colours$.

The approach to take is that we must toggle the **other** $C_1 - C_2$ Kempe Chain of $\{\text{Neighbours}(v_*)\}$, which will be 100% successful. Then color v_* the color of its neighbour that was toggled.
Note: A solution will always exist because our graph is planar. The Kempe Chains act as barriers in the graph, making it impossible for every vertex of $\{\text{Neighbours}(v_*)\}$ to have a Kempe Chain with one another. hence, there exists at least one vertex whose Kempe Chain can be toggled.
The below image shows why this other Kempe Chain will be successful. *There always exists a Kempe chain that doesn't include two vertices of the neighbours of v_* for Case 2.*

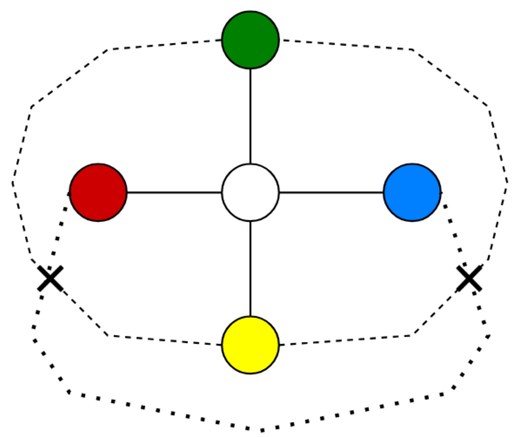


FIGURE 1. A minimal example for the second case. The dotted lines imply a Kempe Chain between two vertices. The Xs denote that it is not possible for Kempe Chains to exists on both vertex pairs.

(3) **Case 3:** $\delta(v_*) = 5$ **and** $\{\mathbf{Col}(\mathbf{Neighbours}(v_*))\} = \mathit{Colours}$**.**
By the Pigeonhole Principle, two vertices will be colored the same color, and the other 3 vertices distinct colors. The goal is the same, to find C_1—C_2 kempe chains that contain only 1 vertex in Neighbours(v_*). Then toggle said vertices so that a free color is available for v_*.
Just like in Case 2, there will always be a neighbouring vertex that doesn't share a Kempe Chain (or so Kempe thought, more on this later). I encourage the unfamiliar reader to draw figures like 1 and 2 to see how there will always be an opening after reading.
(a) Toggling Kempe Chain of **distinctly** colored neighbours.
We first check if we can toggle Kempe chains of distinctly colored neighbours. Doing so frees up a color for v_* in the most convenient manner.
Suppose that there was no Kempe Chain between 2 neighbouring vertices of v_* of different colors C_1—C_2. This means that toggling the C_1—C_2 kempe chain at a vertex colored C_1 or C_2 frees up C_1 or C_2, respectively. Then color v_* the color of the vertex that was toggled for its C_1—C_2 kempe chain.
(b) Toggling (2) Kempe Chains of **undistinct** colored neighbours.
Suppose that there existed Kempe Chains between all distinctly colored neighbours of v_*. When this case occurs, there is a direct implication that makes it

possible for us to toggle the Kempe chains of the undistinctly colored neighbours v_{n1} and v_{n2} **at** v_{n1} **and** v_{n2}. We have to toggle them both in order to swap their colors with something else and free up their current color.

Let $\text{Col}(v_{n1}, v_{n2})$ be $C_{undistinct}$.

I know that this may not be intuitive as it is not displayed, but it turns out that there do not exist kempe chains between v_{n1} and a distinctly colored neighbour v_{d1} and for v_{n2} and a distinctly colored neighbour v_{d2}, where $v_{d1} \neq v_{d2}$. Note that this is only true when there are Kempe chains between all distinctly colored vertices.

So, toggle the $\text{Col}(v_{n1})$—$\text{Col}(v_{d1})$ and $\text{Col}(v_{n2})$—$\text{Col}(v_{d2})$ kempe chains **at** v_{n1} **and** v_{n2}; not at v_{d1}, v_{d3}, as it may not be possible... think with yourself. So, the original color $C_{undistinct}$ is now free and after Kempe chain toggling, v_* may be colored $C_{undistinct}$. The result is that there will be two pairs of neighbours that are colored the same color, and 1 uniquely colored neighbour of v_*.

Similiar to Case 2, we show a figure below highlighting how it is not possible for Kempe chains to exist between every two neighbours of v_*.

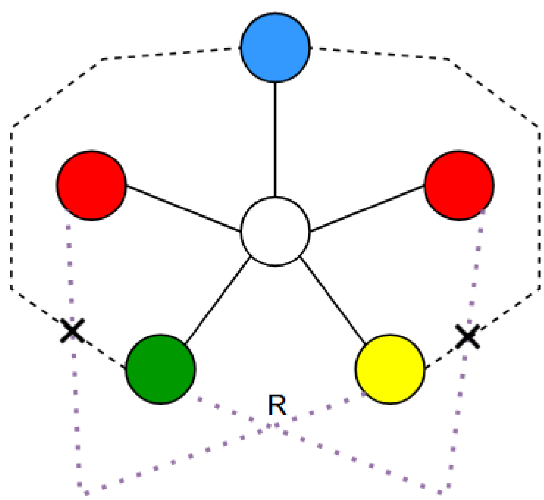


FIGURE 2. Example for Case 3. Kempe chains are mutually exclusive such that it is always possible to toggle some Kempe chains to reduce the cardinality of the neighbouring vertices' color set to 3. Then color v_* the remaining color. The idea is that if there were Kempe chains from the distinctly colored neighbours, then there would be no Kempe chains between the nondistinctly colored vertices with another neighbour, and vice versa. The R signifies an R colored vertex that allows two kempe chains to cross for case b.

This completes the proof of explanation for all possible neighbour cases. Therefore, a connected non-null simple planar graph G with one uncolored vertex v_* whose degree ≤ 5 with all other vertices 4-colored, is 4-colorable by the Kempe Method.

$\square$

Now that the 5-case Kempe Method has been explained. Let us explain how this is used to prove 4CT as Kempe thought, then we will show the countereaxmple created by Heawood.

1.4. **Kempe's Inductive Proof (4CT).**

Theorem 1.8. *(*Main Inductive Algorithm using Kempe Chains)*

Let G be a non-null simple connected planar graph. Ignoring the Heawood counterexample, Kempe's method provides a proof of the 4CT. Assume that the 5-case Kempe chain method which was explained above is valid.

Proof. (By Induction)

Construction 1.9. $\exists$ *a vertex with degree* ≤ 5 *in* G*, i.e.,* G *is 5-degenerate. Let us remove this vertex to form a non-null connected, simple planar induced subgraph* G'*.* $\exists$ *a vertex with degree* ≤ 5 *in* G' *(*G *is 5-degenerate) as well. We recursively find and delete this vertex with degree* ≤ 5 *in all subsequent subgraphs* G'_i *until* G' *is null. Then we repopulate the graph in the reverse order of deletion and apply the 5-case Kempe method for each reinjected node until the graph* G *is reformed and now 4 colored.*

The deletion and reinjection order is like a stack data structure.

Inductive Proof: Let G be a connected non-null simple planar graph. $\text{AvgDeg}(G) \leq 5$.

Let $P(n)$ be the proposition that every connected simple planar graph with $n > 0$ vertices is 4-colorable.

Base case: $P(1)$ is true since a single vertex can be coloured with 1 colour. Trivial that $1 < 4$.

Inductive step: Assume that $P(k)$ is true for some connected simple, planar graph G with k vertices. Add a vertex v_{new} to G and connect it with some vertices $v \in V(G)$ to form $\mathcal{G}$ such that $\mathcal{G}$ is connected, simple, and planar.

$P(k+1)$**:**

- If $\text{degree}(v_{new}) \leq 5$ in $\mathcal{G}$. Color $\mathcal{G}$ with 4-colors by the inductive hypothesis. Then apply the 5-case Kempe method to color (v_{new}) so that $\mathcal{G}$ is 4-colored.
- If $\text{degree}(v_{new}) > 5$ in $\mathcal{G}$. Uncolor the graph $\mathcal{G}$. Then, recursively remove the vertices with degree ≤ 5, as mentioned above. Then repopulate the vertices and color each vertex along the way using the case-by-case Kempe method. Now $\mathcal{G}$ is 4-colored. And the inductive hypothesis is true.

So $P(k+1)$ is true. And by *Mathematical Induction*, Kempe's method proves that G is 4-colorable and therefore 4CT.

□

This is the flawed proof. One can see why this is very convincing, but it is flawed and disproved with the following counterexample. This counterexample was generated at one of the seminars thanks to Prof. Nehaniv, Amena, and Hanna in January 2026.

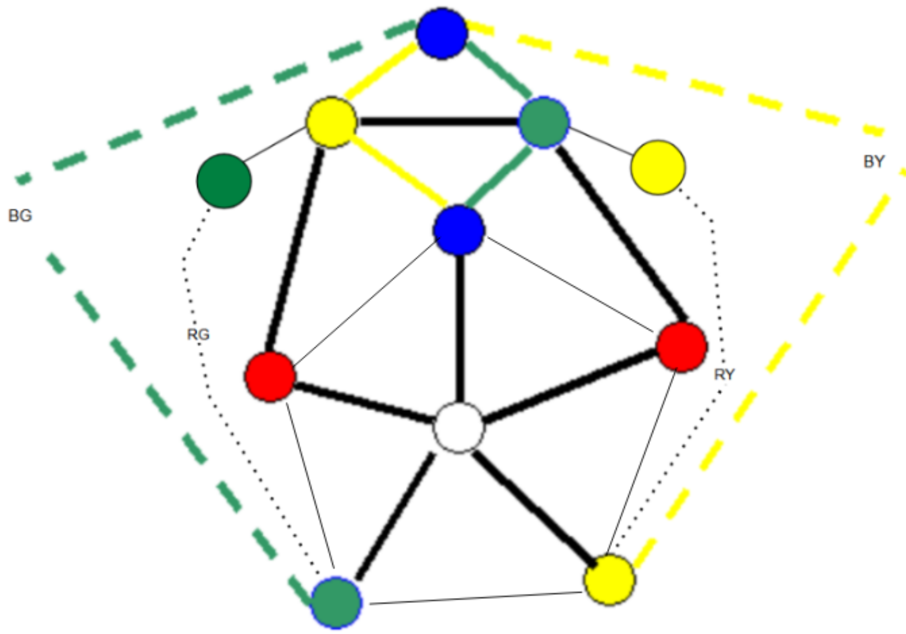


FIGURE 3. Counterexample for Kempe's method formulated Percy Heawood and our team of Prof. Nehaniv, Amena, Hanna, and I. We adapted Shai Simonson's original photo to produce a more meaningful counterexample! [3]. Upon trial, one will see that is never possible to free up a color for the uncolored vertex v_* by toggling Kempe Chains of its neighbours. This is because one will see that toggling the R-Y Kempe chain at the left R vertex, creates an R-G Kempe chain at the right R vertex. Image readapted from from Shai Simonson [6].

1.5. **5-Color Theorem.**

The result of the failed Kempe Method on 4CT is actually the 5CT.

Theorem 1.10. *(5-Color Theorem)*
Any non-null connected planar, simple graph is 5-colorable.

Proof. (Adaption to the 4 color Kempe Chain Method)
Use the Kempe Chain Method described above. But, when the counterexample case occurs, simply color the uncolored vertex (previously denoted v_*) the fifth color. Since the 5th colored vertex will only be adjacent to neighbours colored with the original 4 colors, this method is decoupled and intuitive. Technically speaking, the 5th color is only to satisfy the counterexample case. But, the idea and thing about coloring is like saving money, we basically just introduced more possibilities/probabilities. □

I implemented the Flawed Kempe method and 5-Color Theorem adaptation in code, which can be seen in the listed directory in the Abstract . The code is the important back spine of this paper

2. Kempe Chain for Hadwiger's Conjecture

2.1. Hadwiger's Conjecture and Important Works.

Definition 2.1. *(Hadwiger's Conjecture)*

The conjecture was made by Hugo Hadwiger in 1943.

It states that if a simple graph G has no K_t minor then its chromatic number satisfies $\chi(G) < t$.

From [13], *"It is known to be true for $1 \leq t \leq 6$. The conjecture is a generalization of the Four Color Theorem and is considered to be one of the most important and challenging open problems in the field."*

Insight: A graph with a K_t minor needs t colors. So Hadwiger's conjecture asks if a graph with no K_t minor should require less than t colors, or is $(t-1)$-colorable.

Theorem 2.2. *(Wagner's Theorem 1937A)*

A graph G is planar if and only if G does not contain K_5 or $K_{3,3}$ as a graph minor. Definition helped by [9].

Definition 2.3. *(Clique-Sum)*

From [11], *a clique of a graph G is an induced subgraph of G that is complete.*

Clique-sum:

Directly quoted and modified from [12]. *For two graphs G and H each containing cliques of equal size, the clique-sum of G and H is formed from their disjoint union by identifying pairs of vertices in these two cliques to form a single shared clique. A k-clique-sum is a clique-sum in which both cliques have exactly k vertices.*

Theorem 2.4. *(Wagner's Theorem 1937B)*

From [2], *a graph has no K_5 minor if and only if it can be constructed from planar graphs and V_8 using* 0-, 1-, 2-, *and* 3-*sums.*

The author realized that Wagner's theorem only requires 1-, 2-, 3- sums (and V8) because K_4 can be generated by 3-sums, but a K_n can not be generated by [1, n-1]-sums for $n = [2, 3]$.

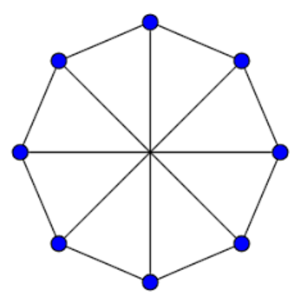


FIGURE 4. V_8 or the Wagner graph. [10].

2.2. Connecting Hadwiger's Conjecture and Kempe Chains.

The aforementioned 1.10 using Kempe Chains implied that a simple planar graph G is 5-colorable, or has a chromatic number satisfying $\chi(G) < 6$. By Wagner's Theorem 1937A 2.2, a graph G is planar if and only if it contains neither K_5 nor $K_{3,3}$ as a graph minor. As a result, 1.10 implies that graphs without K_5 or $K_{3,3}$ minors are 5-colorable.

That was the Kempe chain approach to 4CT/5CT. Now, let us show the Kempe chain approach to Hadwiger's Conjecture for $t = 5$ (no K_5 minor). This will help us grow an understanding on Hadwiger's Conjecture and kempe chains.

Graphs with no K_5 minor are 4-colorable using Kempe Chains: First, we assume that the 4CT is true. By Wagner's Theorem 1937B 2.4, a graph G with no K_5 minor can be thought of as the joining of planar subgraphs and V_8 subgraphs(or $K_{3,3}$) along 2-sums. Since a planar graph is 4-colorable by 4CT, and V_8 is 3-colorable, then the joining of these two subgraphs to form a graph without a K_5 minor G therefore must be 4-colorable.

Suppose we have a planar graph and V_8; 4-color the planar graph and 3-color V_8. Then, before joining the two graphs to form a graph G with no K_5 minor, identify the colors of the clique-sum vertices. If the 2-cliques on both subgraphs of G have corresponding vertices of the same color, then G will be colored properly and therefore 4-colorable. If the colors on corresponding vertices of the clique are differently colored simply *toggle* the kempe chains on these vertices one at the time or at the same time if possible, so that the clique sum vertices on both the planar subgraph and V_8 are the same. Note: one can toggle the kempe chain of vertices on either subgraph.

As we keep building the graph G, we do the same thing. Even when the subgraphs H and M are now just graphs with no K_5 minor each themselves and we effectively just apply Wagner's Theorem 1937B 2.4 over and over, we can still do the same Kempe chain toggling for K_1, K_2, K_3(, and I'm pretty sure K_4) clique-sums.

> The logic is that suppose we didn't toggle kempe chains, but instead we swapped certain colors of the *entire* subgraph to ensure matching corresponding clique sum vertices, wouldn't toggling kempe chains be the same thing?!
> If this is not clear, simply said: we can always make the "glueing vertices" the same color so that the joined graph is properly colored.
> So, kempe chains can be said to help prove Hadwiger's conjecture for $t = 5$: graphs with no K_5 minor are 4-colorable!

Let us now develop our ideas with the seed planted and reap what we sow.

Connecting Kempe Chains to solve Hadwiger's Conjecture is not as simple as the above makes it seem. So, starting from the basics, the starting idea of connecting Kempe chains with Hadwiger's Conjecture 2.1 will be done through the simple difference between 4CT/5CT and Hadwiger's Conjecture for $t = 5$ (no K_5 minor), that is, **the $K_{3,3}$ minor**.

2.3. **Kempe Method - Hadwiger Conjecture Counterexamples.**

I first tried to see if I could use the Kempe chain method to show that graphs with no K_t minor for $t = 5$ are 5-colorable; directly translating Kempe's 5CT 1.10(which works) to Hadwiger's conjecture. A counterexample to disprove the ability to use Kempe's method for $HC(5)$ was generated; I am pretty sure simpler counterexamples exist.

For the case seen below with 5 neighbors, each of different colors, of v_*, The introduction of $K_{3,3}$ in the form of V_8 made Kempe's case by case method implausible. It is not possible to successfully toggle any Kempe chain as all neighboring vertices form Kempe chains with each other via $K_{3,3}$.

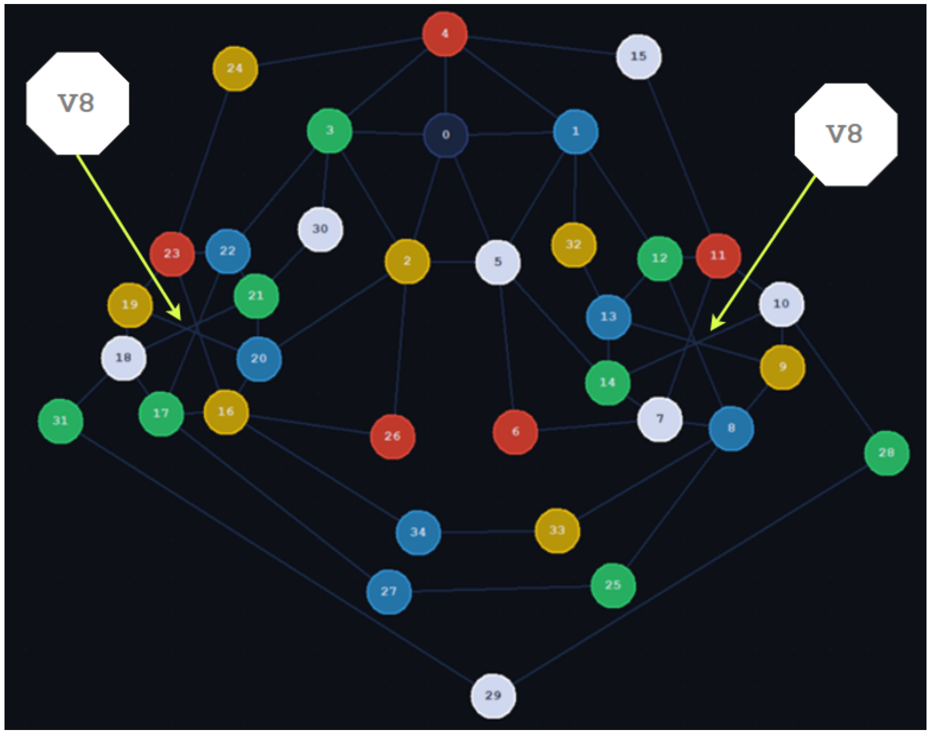


FIGURE 5. Another counterexample for Kempe's method for $HC(5)$. An easier counterexample is highly likely to be possible to demonstrate for 5 colors, but this will work.

2.3.1. *When Nonplanarities exist, Kempe's method does not apply.*

The conclusion is that when nonplanarities exist, Kempe's method does not apply.

This can be visualized for trying to prove HC(5) (graphs with no K_5 minor are 4 colorable) using Kempe's method. We already know it fails, even for just planar graphs, but the below example shows the simplest reason why non planarities simply don't apply with kempe chains; they create too much of them.

Below v_* has 4 neighbours that complete the entire coloring set of cardinality 4. The introduction of $K_{3,3}$ in the form of V_8 makes Kempe's method implausible. It is not possible to successfully toggle a Kempe chain, as all vertices form Kempe chains with one another.

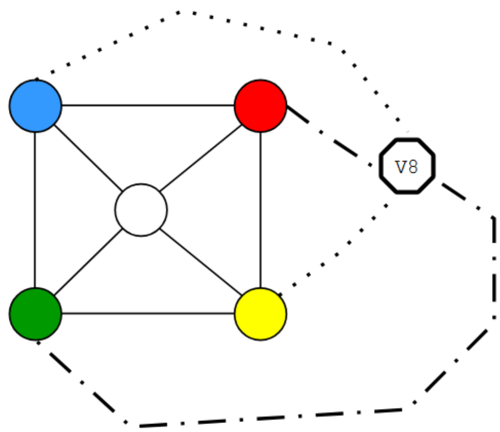


FIGURE 6. Counterexample for Kempe's method for $HC(5)$. Adding non-planarity via $K_{3,3}$ create a Kempe chain between all neighbours.

2.4. **The Modified Kempe Algorithm - Beginning.**

Since we know that Kempe's original method doesn't work to solve Hadwiger's Conjecture due to non planarities, we had to create a modified version. This modified version still uses Kempe Chains, as we know that is intuitive. But, the algorithm is different. This modified Kempe Algorithm allows the Kempe method/Kempe chains to never die, allowing a potential application into Hadwiger's Conjecture.

In this modified Kempe algorithm, the idea of Kempe chains and the case-by-case Kempe method seen in 1.3 is exactly the same, the only difference is in the algorithm/implementation of this method in coloring, which was previously 1.8.

The modified algorithm allows $K_{3,3}$ (and I infer all non planar) minors to be allowed. The modified version strives to target Hadwiger's Conjecture. Originally, the goal of the modified version was to investigate and show that graphs G without K_5, K_6 minors are potentially 5, 6-colorable, respectively. This is 1 color weaker than Hadwiger's Conjecture, but it actually turns out that the modified version may directly apply to Hadwiger's Conjecture. More on this later.

Let us first begin by presenting important definitions and theorems. This may be a bit boring and repetitive, but it is necessary to know what we refer to when formally introducing the Modified method. Feel free to skip to the formal explanation of the Modified Kempe method if one desires, as the below is important but fairly simple.

Definition 2.5. *($k-$regular Graph)*

A k-regular graph is a graph in which every vertex has the same number of neighbors k, that is, all vertices have the same degree k. Therefore, a k-regular graph is k-degenerate and k+1 colourable.

Definition 2.6. *(Maximally K_n Graph)*

A maximally K_n graph G is a graph for which every vertex $v \in V(G)$ has at least $n-1$ neighbours and each unique neighbouring set of size $n-1$ vertices plus v form a K_n minor $\forall v \in G$.

Lemma 2.7. *(Clique-sum minors)*

For a graph G formed by the clique sum of graphs M and N, which have no K_m and K_n minor, respectively, G has no max(m, n) minor. And it follows that the degeneracy may be determined for $m, n \leq 7$, following the conditions of 2.9

Proof. Trivial. □

Theorem 2.8. *(Forbidden Minors $\implies$ Edge-bounds (one-way))*
*For $2 \leq t \leq 7$ and for all *$v \geq t - 2$* (this is important so that $e > 0$) every v-vertex finite graph G with no K_t minor satisfies $e \leq (t-2)v - (t-1)(t-2)/2$.*

Proof. Let G be a graph with no K_t minor. The maximum number of edges in G occurs when G is maximally K_{t-1}, see definition (2.6).

So let's form the "base" of our maximally K_{t-1} graph G. By the Handshaking lemma 1.4, we know that K_{t-1} has $\frac{(t-1)(t-2)}{2}$ edges. Every added vertex will connect to $t-2$ vertices so that G is maximally K_{t-1}. So, the total number of edges in a maximally K_n graph is:

$e =$ number of edges in $K_{t-1}+$ (number of vertices)(number of added edges for maximal K_{t-1})

$$\implies e = (t-1)(t-2)/2 + [v - (t-1)](t-2)$$

where $[v-(t-1)]$ is the number of added vertices on top of K_{t-1}, and $(t-2)$ is the number of added edges for each added vertex.

$$\implies e = (t-1)(t-2)/2 + [v - (t-1)](t-2)$$

$$e = (t-1)(t-2)/2 + (t-2)(v) – (t-2)(t-1)$$

$$e = -(t-1)(t-2)/2 + (t-2)(v)$$

$$\implies e = (t-2)(v) - (t-1)(t-2)/2, e \in \mathbb{W}$$

So the number of edges in a maximally K_{t-1} graph is $(t-2)(v) - (t-1)(t-2)/2$. Or, for $2 \leq t \leq 7$ and all $v \geq t-2$, every v-vertex finite graph G with no K_t minor satisfies:

$$e \leq (t-2)(v) - (t-1)(t-2)/2$$

(For $2 \leq t \leq 7$)

(1) ($t = 1$). (A graph with no K_1 minor is a null-graph and $e = 0$.)
(2) $t = 2$. A graph with no K_2 minor up to 1 vertex and $e \leq 0$ is satisfied.
(3) $t = 3$. A graph with no K_3 minor could be thought of as a tree and $e \leq v - 1$ is satisfied.
(4) $t = 4$. A graph with no K_4 minor could be thought of as a series-parallel graph and $e \leq 2v - 3$ is satisfied.
(5) $t = 5$. A graph with no K_5 minor could be thought of as a clique sum of planar and $V8$ graphs and $e \leq 3v - 6$ is satisfied.
(6) $t = 6$. A graph with no K_6 minor has $e \leq 4v - 10$ is satisfied.
(7) $t = 7$. A graph with no K_7 minor has $e \leq 5v - 15$ is satisfied.

□

Theorem 2.9. *(Edge-bounds $\implies$ Degeneracy (one-way))*

For any non-null finite graph G with an edge-bound, we can determine a bound on the average degree and the degeneracy of G.

Therefore, for $2 \le t \le 7$ every v-vertex graph G with no K_t minor has a degeneracy bound using the upper edge bounds 2.8.

Proof. Suppose that there is an edge bound $E(v)$ on a graph G with $v > 0$ vertices such that $e \le E(v)$. Consider for $2 \le t \le 7$, v-vertex graphs G with no K_t minor have $e \le (t-2)v - (t-1)(t-2)/2$ from 2.8.

Let us use these edge bounds to determine the AvgDeg and degeneracy of these graphs.

$$\text{Average Degree}(G) = \frac{2e}{v}$$

By Handshaking lemma 1.4.

$$e \le (t-2)v - (t-1)(t-2)/2$$

$$\implies \text{AvgDeg}(G) \le \frac{2(t-2)v - (t-1)(t-2)}{v}$$

$$\implies \text{AvgDeg}(G) \le 2(t-2) - \frac{(t-1)(t-2)}{v}$$

$$\implies \text{AvgDeg}(G) \le (t-2)(2 - \frac{(t-1)}{v})$$

Remember that $v > 0$. An upper bound exists when $v \to \infty$. v is finite; so let v approach infinite, but never equal.

$$\implies \text{AvgDeg}(G) \le (t-2)(2 - 0^+)$$

$$\implies \text{AvgDeg}(G) \le (t-2)(2^-) \in \mathbb{R}$$

$$\therefore \text{AvgDeg}(G) < 2(t-2)$$

For $t = 2$, refer to $\text{AvgDeg}(G) \le (t-2)(2^-) = \mathbf{0}$ since $t - 2 = 0$

And so for non-null finite graphs G with no K_t minor:

- $t = 2$**:** $\text{AvgDeg}(G) = 0$. This tell us that there will always exist a vertex with degree 0.
- $3 \le t \le 7$**:** $\text{AvgDeg}(G) < 2(t-2)$. This tell us that there will always exist a vertex with degree $\le 2(t-2) - 1$.

G is $(2t-5)$-degenerate for $3 \le t \le 7$ and 0-degenerate for $t = 2$.

(For $2 \leq t \leq 7$)

(1) $t = 2$. A graph without K_2 minor has up to 1 vertex and Average Degree $= 0$ is satisfied.
(2) $t = 3$. A graph without K_3 minor could be thought of as a tree and Average Degree < 2 is satisfied. G is 1-degenerate.
(3) $t = 4$. A graph with no K_4 minor could be thought of as a series-parallel graph and Average Degree < 4 is satisfied. G is 3-degenerate.
(4) $t = 5$. A graph with no K_5 minor could be thought of as a clique summing planar and $V8$ graphs and Average Degree < 6 is satisfied. G is 5-degenerate.
(5) $t = 6$. A graph with no K_6 minor has Average Degree < 8 is satisfied. G is 7-degenerate.
(6) $t = 7$. A graph with no K_7 minor has $e < 10$ is satisfied. G is 9-degenerate.

□

In 1.5, we saw that the degeneracy of a planar graph was 5. Here we see that a graph with no K_5 minor is also 5-degenerate. Since $K_{3,3}$'s largest minor is K_2 or K_3 for V_8, this means that a planar graph's largest minor is also K_4. Hence a graph with no K_5 minor and a planar graph have the same degeneracy and maximum edgebound. This shows the relationship!

It is much easier and intuitive to obtain the degeneracy of general graph this way then compared to Euler's Formula in 1.5.

2.4.1. *Upper bound on t.*

The above theorems, 2.8 and 2.9, apply only to $2 \leq t \leq 7$.

For $t = 8$ and higher, they are well studied to fail as the edge-bound and hence the average degree-bound are exceeded. Mader showed that $K_{2,2,2,2,2}$ which has no K_8 minor does not satisfy bounds for $t = 8$. The reason why $K_{2,2,2,2,2}$, which not only has no K_8 minor, but also has no K_6 minor fails becomes obvious and intuitive upon investigation.

$$e \leq (t-2)(v) - (t-1)(t-2)/2$$

So, obviously we know that a graph with no K_t minor has a maximum number of edges if the graph was maximally K_{t-1} or $K_{\underbrace{1,1,\ldots 1}_{\text{(t-1) 1s}}}$, but we know, by the definition, that these graphs fit the inequality. Looking at the equation, we see that adding more vertices, we increase our edge bound so that won't work.

So we must artifically increase t, while keeping v low. What we found out is that for the same v at $t = t_0$ and $t = t_1 = t_0 + 2$, the edge bound actually decreases. Try it yourself using the formula

E.g. a graph with no K_6 minor has $e \leq 4v - 10$; a graph with no K_8 minor has $e \leq 6v - 21$. For v = 5, $e \leq 10, 9$ respectively. So, we can artificially decrease the edge bound this way, and finding a way to disprove this equation for **higher** t.

In fact K_5 or $K_{1,1,1,1,1}$ which has no K_8 minor and 10 edges disproves the edge bound for $t = 8$, and we see that Mader's $K_{2,2,2,2,2}$ follows as well!

Theorem 2.10. *(Mader's Theorem; proved by me)*

"If the average degree of G (which has v vertices) is at least 2^{t-2} then G has a K_t minor." Theorem statement obtained from [4].

Proof. 2.9 showed that for $2 \leq t \leq 7$ and all $v \geq t-2$, every v-vertex graph G with no K_t minor has a degeneracy (or average degree) bound.

Now, we are basically trying to prove the inverse, or go from $\leq$ to $>$. For $t = 2$, $3 \leq t \leq 7$, the upper bound on the average-degree for which there is no K_t minor is AvgDeg $= 0$ and Average Degree$(G) < 2(t-2)$, respectively. As a result, we know that exceeding this implies a K_t minor.

And for $t > 7$ though, we know that our edge-bound and the average degree bound fails, such that there are graphs with no K_t minor that exceed Average Degree$(G) < 2(t-2)$. This failure actually does not matter in this proof, contrary to what I first thought. The average degree bound does not actually care on whether it is successful or not. It only matters if it **fails**. Because when it fails, then we go from a maximally K_{t-1} graph to a K_t graph.

Therefore, any graph that does not follow our bound from 2.9 has a K_t minor. $\implies$ If Average Degree$(G) >= 2(t-2)$, we know that the graph as a K_t minor.

Since $2^{t-2} \geq 2(t-2) \forall t > 0$ and $2^{t-2} > 0$ then the inequality applies for all $t >= 0$.

$\implies$ The theorem is proved. Please read the blue text for more insight on why it applies for all $t > 0$.

The inequality can be shown graphically. See the visual below.

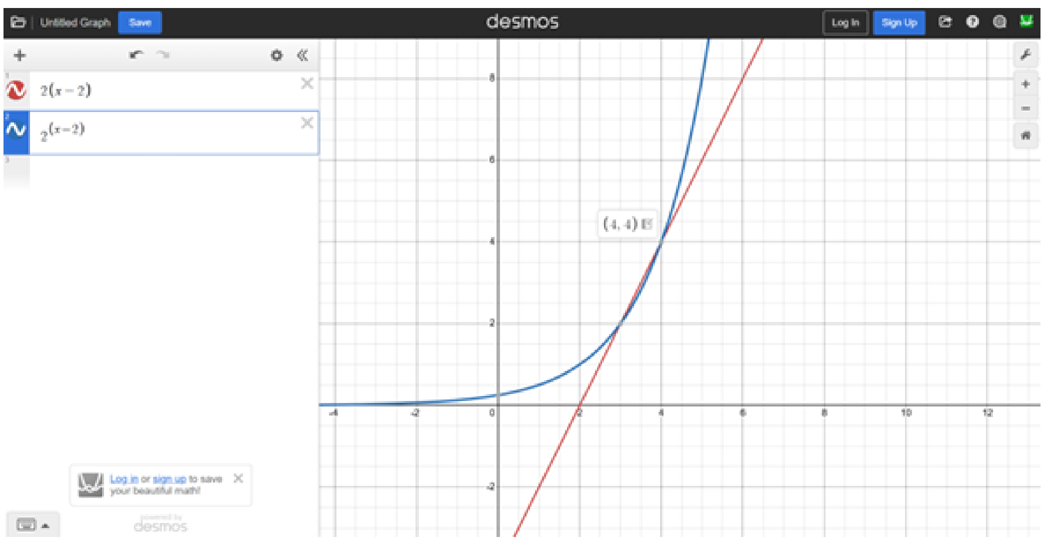


FIGURE 7. 2^{t-2} is strictly positive and $2^{t-2} \geq 2(t-2) \forall t$. So if the average degree of G is at least 2^{t-2} then G has a K_t minor.

$\square$

3. Modified Kempe Chain Algorithm Solves Hadwiger's Conjecture?

Everything below may come across as informal, it's because this is a proposition.

3.1. **The Modified Kempe's Method.**

We created the modified method to try to show that simple/loopless graphs with no K_t minor are t-colorable (1 color weaker than HC). Assume that $2 \leq t \leq 7$ for now, since we know their degeneracies, by 2.9, but the modified method may possibly work for higher t as well.

Note: This method has been programmed. Please contact the email in the abstract .

Construction 3.1. *(The Modified Kempe Chain Algorithm)*
Given a loopless graph G with no K_t minor, we know it is k-degenerate (for $2 \leq t \leq 7$). So there exists a vertex v_* with degree $\leq k$.

1. *Find v_* and color its neighbours using Kempe chains (Kempe's method). If its neighbours are not colorable, find another v_**
2. *Color v_* using Kempe chains (Kempe's method). v_* may or may not be already colored.*
3. *Once v_* is colored properly, remove it and its edges from G to form G'.*
4. *Repeat until all vertices in G' are colored. G' will not finish as a null graph.*
5. *Add back all deleted vertices v_* at the same time. For all $v_{*,i}$, check if there is a coloring conflict. If so, color $v_{*,i}$ the t^{th} color. Note: coloring conflicts occur due to Kempe chain toggles that that inferred a non-existent $v_{*,i}$.*

Note: I originally made two assumptions about the modified Kempe's method out of my delusion, see 6.1 in the Appendix for more insight.

I think the most important thing about this modified algorithm is the fact that we assume that $(t-1)$ colors, remember that the t_{th} color is for coloring conflicts, is sufficient for coloring any vertex with up to k neighbours. This "assumption" should be investigated further.

As I thought more about coloring, I realized that the timeline in graph coloring was as follows: 4CT (1850s-1970s), Kempe (1879), then Hadwiger's Conjecture (1943). What surprised me was how Hadwiger's Conjecture is so recent, as it is a very intuitive problem (that I think is true). But what I truly realized was that the number **4** for planar graphs and Kempe's original method was originally just from observation. Suppose that Kempe tried his method before 4CT was popularized, wouldn't the number 4 just be considered a random number with no trend/meaning? What I am going with here is that I thought that the modified kempe chain method would be 1 color weaker than Hadwiger's Conjecture. But then I went back in time, *but in thought*, and I realized that why can't the Modified Kempe's Method provide a gateway into solving Hadwiger's Conjecture directly?

So I overcame this mental hurdle and realized that since the modified method overcomes non-planarities, there's no reason why it can't be a computer algorithm that works in practice to prove Hadwiger's Conjecture or at least 1 color weaker than Hadwiger's Conjecture.

- **The scalability of Kempe's Chains:** When I originally thought about Kempe Chains, I thought that it was directly coupled with his flawed 4CT and subsequent 5CT for planar graphs. So as I scaled through Hadwiger's Conjecture, I always thought that the Kempe method would be at least 1 color weaker than HC, since its only success was for 5CT on **planar** graphs. But I slowly realized that the Modified Kempe's method, which breaks free from the Heawood counterexample, may actually work prove Hadwiger's Conjecture. An exact 1-to-1 in chromatic number! The modified Kempe method is not restricted to planarity, and my hope is that this algorithm can be used in practice for coloring random graphs. I have some example graphs which convince me to believe this algorithm CAN be a 1-to-1 chromatic coloring with HC...

Why?: Well, when Kempe first introduced his flawed method, all planar graphs were mistakenly shown to be 4 colorable. Then, the Heawood counterexample proved it false, and it showed Kempe's method won't work with non-planar graphs, for sure. See 6 and 5.

Well, if the modified Kempe's method can bypass the troubles that non-planarity introduces in Kempe's method (since non planarities can create Kempe chains between all neighbours), and it passes the Heawood counterexample, I have a nice pipeline/reason to believe that this modified method which: works with non-planarities (and can be scaled with HC), and bypasses Heawood's counterexample (due to that nature of the method), is actually directly trying to solve HC.

The transition: Kempe originally showed using Kempe Chains that planar graphs are 4-colorable, excepting the Heawood counterexample. Since the modified Method includes non planarities and bypasses the counterexample, it is saying that it bypasses the one thing that made Kempe's original proof flawed, that is: too many Kempe Chains (A.K.A non planrities). So, practically, the modified Kempe method would show that graphs with no K_5 minor are 4-colorable. What's amazing here is that the Kempe Chain method of Kempe actually gave us so much insight on Hadwiger's Conjecture. BECAUSE, even if Kempe didn't know 4CT existed, he would've gotten the number 4.

> The modified method bypasses planarities through the 4th dimension! Since non-planarities represent 3D!!!

4. Lower bound on the number of $\leq k$-degreed vertices in a k-degenerate graph

4.1. **Critical's Theorem.** I originally thought that for a k-degenerate graph G, I could show that there are at least k+1 vertices with degree $\leq$ k. This seemed very important for me to prove the modified Kempe's Method, I don't think it's important anymore. Instead, it seems that for $2 \leq t \leq 3$, $4 \leq t \leq 7$, graphs with no K_t minor, and are k-degenerate, have at least $(k+1)$, $(t-1)$ vertices with degree $\geq k$, respectively.

I will first begin with my failure and then with the counterexamples that bring me to my new conclusion.

Theorem 4.1. *For $2 \leq t \leq 7$ and all $v \geq t-2$, every v-vertex finite graph G with no K_t minor has a determinate $k-$ degeneracy (2.9) and it follows that there exists a subset of vertices $V' \subseteq V(G)$ such that $|V'| \geq k+1$ and every $v \in V'$ satisfies having a degree $\leq k$. Formally:*

$$|\{v \in V(G) \mid \deg(v) \leq k\}| \geq k+1$$

for a k-degenerate graph G without K_t minor.

Proof. (Disproof with counterexamples)

Construction 4.2. *K_n is $n-1$-degenerate, or, each of the n vertices has $n-1$ vertices. Therefore, a k-degenerate graph cannot have a K_{k+2} minor, since the degree of each vertex exceeds the degeneracy by 1.*

Next, we ask ourself: can a k-degenerate graph have a K_{k+1} minor. Obviously, K_{k+1} is k-degenerate. So, there are some graphs with a K_{k+1} minor that are k-degenerate. But do these graphs satisfy:

$$|\{v \in V(G) \mid \deg(v) \leq k\}| \geq k+1?$$

Suppose that we have K_{k+1}, with degree$(V) = k$. Then add a vertex v and join v with k vertices $v_1, v_2, \ldots, v_k$ in K_{k+1}. We are 1 edge away from K_{k+2}, which is not k-degenerate. It follows that there are k vertices now with degrees $k+1$ and only 2 vertices with degree k, the vertex v and v_{k+1}. Therefore, if a graph has a K_{k+1} minor, the condition fails.

If a graph has a K_k minor, all its vertices have degree $k-1$. Therefore adding another vertex and connecting to $k-1$ vertices of K_k (not k vertices, otherwise a K_{k+1} minor is formed), a graph where all $k+1$ vertices have degree $\leq k$ is developed. This satisfies the condition.

When we translating this idea with our theorem 2.9 *we start making sense of an idea.* 2.9 *says that the degeneracy of a v-vertex graph G with $v \geq t-2$ with no K_t minor for $t \leq 7$ has a degeneracy bound.*

Generalising the trend is as follows. A maximally K_n graph with a k-degeneracy limitation and at least k+1 vertices of degree less than or equal to k only exists if $k \geq n$ or $k \geq t-1 \implies k+1 \geq t$

(1) *$t = 1$. A graph without K_1 minor is 0-degenerate*
(2) *$t = 2$. A graph without K_2 minor is 0-degenerate*
(3) *$t = 3$. A graph without K_3 minor is 1-degenerate.*
(4) *$t = 4$. A graph without K_4 minor is 3-degenerate.*
(5) *$t = 5$. A graph without K_5 minor is 5-degenerate.*
(6) *$t = 6$. A graph without K_6 minor is 7-degenerate.*
(7) *$t = 7$. A graph without K_7 minor is 9-degenerate.*

Notice how degeneracy increases by 2 each time?

For $v \leq k+1$, it is obvious that all v vertices will have a degree less than or equal to k, or be k-degenerate. This is because the max degree of a v-vertex graph is v-1. Even more so with minor restrictions, decreasing the edge and degree count!

Specifically, we see that for $4 \leq t \leq 7$, graphs that have no k_{t+1} minor are k-degenerate graphs. This follows our condition of the existence of at least $k+1$ vertices with degree $\leq k$. For $t=3$, graphs with no K_3 minor are 1-degenerate (or trees). Let's apply the same experiment. I'll simply substitute k=1 into the above statement.

Assume we have $K_{k+1=2}$, with degree$(V) = k = 1$. Then add a vertex v and join v with $k=1$ vertices v_1 in $K_{k+1=2}$. We are 1 edge away from $K_{k+2=3}$, which is not k=1-degenerate. It follows that there are $k=1$ vertices now with degrees $k+1=2$ and only 2 vertices with degree $k=1$, the vertex v and $v_{k+1=1}$. As the reader can see, by virtue of the fact that joining K_n to a vertex v at $n-1$ vertices, we always generate 2 vertices of degree k and n-1 vertices of degree k+1. This generally wouldn't satisfy the condition stated in the theorem. But, a graph with no K_3 being 1-degenerate (difference greater than 1) actually just satisfied our condition, as there are $k+1=2$ vertices for with degree $\leq k$.

For t = 2 $\implies$ $k=0$, we also see that the difference between t and k to be greater than 1. It turns out that this difference also doesn't matter by virtue of the beauty of mathematics. If a non-null graph has no K_2 minor, it is simply a vertex. And it turns out that a vertex has 1 vertex with degree = 0. This satisfies the no K_t minor - k-degeneracy law condition, as there are $\geq k+1 = 1$ vertices with a degree less than or equal to k= 0.

For t = 1, the condition fails, as the graph is null. So there can not be at least 1 vertex.

Therefore, for $2 \leq t \leq 7$, we find that the No-K_t minor —– k-degeneracy condition holds on G, so that it is possible that at least $k+1$ vertices with degree $\leq k$ exist in the graph G.

We know that a k-degenerate graph contains at least 1 vertex with a degree less than or equal to k. A counterexample to the above theorem would be a k-degenerate graph where there are $\leq$ k vertices with a degree less than equal to k.

A maximally K_{t-1} graph has the most edges for a graph with no K_t minor. For $t \leq 7$, this edge-bound is known and is stated in 2.8. We show that all **non null**, finite, connected graphs G with no K_t minor, and thus a k-degeneracy, for $t \leq 7$ and at least k+1 vertices, have at least $k+1$ vertices with $degree \leq k$ by showing a relationship between a maximally K_{t-1} graph.

For $t \leq 7$, let us assume a graph G with no K_t minor, or maximally K_{t-1}. From 2.8:

(1) $t=1$, or maximally K_0: $e=0$.
(2) $t=2$, or maximally K_1: $e=0$.
(3) $t=3$, or maximally K_2: $e=v-1$.
(4) $t=4$, or maximally K_3: $e=2v-3$.
(5) $t=5$, or maximally K_4: $e=3v-6$.
(6) $t=6$, or maximally K_5: $e=4v-10$.
(7) $t=7$, or maximally K_6: $e=5v-15$.

Knowing this edge bound, let's assume a graph G with no K_t minor, and thus a k-degeneracy, for $t \leq 7$. Let it be be maximally K_{t-1}. Assume that there are exactly k+1 vertices $V(G)$ with $degree = k$ and other vertices have a degree k+1.

$$\implies \sum_{v \in V(G)} \deg(v) = k(k+1) + (k+1)[v-(k+1)]$$

$$\implies \sum_{v \in V(G)} \deg(v) = k^2 + k + (k+1)(v-k-1)$$

$$\implies \sum_{v \in V(G)} \deg(v) = (k^2 + k) + (vk - k^2 - k + v - k - 1)$$

$$\implies \sum_{v \in V(G)} \deg(v) = vk + v - k - 1$$

$$\implies \sum_{v \in V(G)} \deg(v) = v(k+1) - k - 1$$

By Handshaking Lemma 1.4

$$\implies 2e = v(k+1) - k - 1$$

$$\implies e = \frac{v(k+1) - k - 1}{2}$$

This implies that the degeneracy must be odd, which is appropriate for $3 \leq t \leq 7$.

Therefore, for graphs with no K_t minor, and thus a determinate degeneracy k... Assuming the graph has exactly k+1 vertices with degree k and all other vertices degree k+1. The number of edges is:

(1) $t = 1, k = 0$ not applicable since the inherent assumption is that there are $k + 1 > 0, k \geq 0$ vertices.
(2) $t = 2, k = 0 \implies e = (v-1)/2 = 0$ for v = 1.
(3) $t = 3, k = 1 \implies e = v - 1$.
(4) $t = 4, k = 3 \implies e = 2v - 2$.
(5) $t = 5, k = 5 \implies e = 3v - 3$.
(6) $t = 6, k = 7 \implies e = 4v - 4$.
(7) $t = 7, k = 9 \implies e = 5v - 5$.

Note: we always use a minimal forbidden minor t and minimal k-degeneracy as that is non trivial $(t, k = \infty \forall G)$ and provides the lowest bound for our edge count.

The results show that the edge count for this graph G is strictly greater than or equal to the edge count for a maximally K_n graph. We know this is not possible as the edge bound above is strict. This implies that there are either:

Case 1:: there are more than k+1 vertices with degree k.
Case 2:: there are less than k+1 vertices with degree less than k and these vertices have a sufficiently low degree such that the edge bound is satisfied.

I was unable to solve case 2... I realized that my approach was flawed. I simply started **drawing** graphs, and I realized the flaw. I found various counterexamples.

I found a counterexample for $t = 5, k = 5$. Assume a maximally K_4 graph. Starting with K_4, each vertex has a degree of 3. If every one of these K_4 vertices had a degree of 6 ($k + 1$ for a k-degenerate graph), that would require $4 * (6 - 3) = 12$ total added edges. Since we notice that every newly added vertex will have 3 edges for a maximally K_4 graph. Therefore, we can add 4 vertices such that there exists only $4 < k + 1 = 6$ vertices with degree less than or equal to $k = 5$.

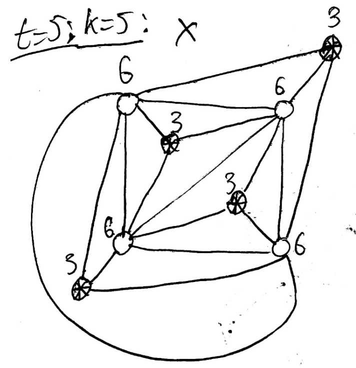
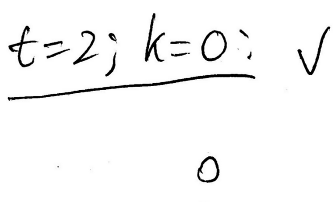
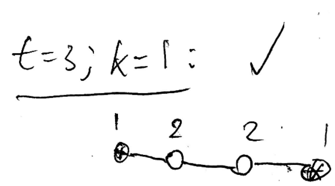


FIGURE 8. This graph with no K_5 minor $\to$ k=5-degenerate is a counterexample (X). It has only has $4 < k+1 = 6$ vertices with degree $\leq k = 5$.

See below for other counterexamples for a graph with no K_t minor for $4 \leq t \leq 7$. For $2 \leq t \leq 3$, the theorem seems to work, but due to the nature that the numbers $0, 1, 2,$ and maybe 3 intrinsically have more meaning/exceptions.



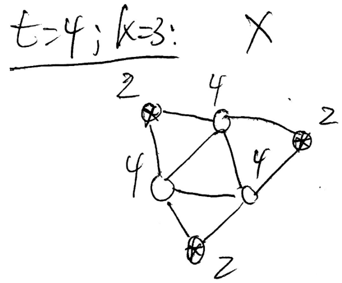
(A) For $t = 2 \to k = 0$, **there are** at least $k+1 = 0$ vertices with degree $\leq k$.



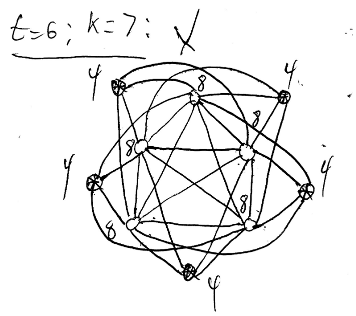
(B) For $t = 3 \to k = 1$, **there are** at least $k+1 = 2$ vertices with degree ≤ 1.

FIGURE 9. Cases ($t = 2, 3$) where the condition SEEMS to apply..

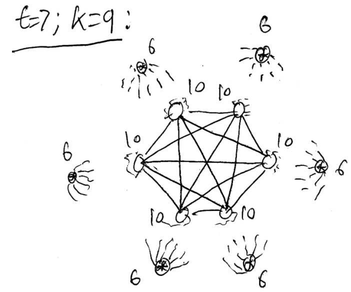
□



(A) Counterexample for $t = 4 \to k = 3$. **There are** $t-1 < k+1$ vertices with degree $\leq k$.



(B) Counterexample for $t = 6 \to k = 7$. **There are** $t-1 < k+1$ vertices with degree $\leq k$.



(C) Counterexample for $t = 7 \to k = 9$. **There are** $t-1 < k+1$ vertices with degree $\leq k$.

FIGURE 10. Cases ($t = 4, 6, 7$) where a counterexample for the condition is found ($t = 5$ above).

5. Conclusion

I need to come up with some examples (or counterexamples) that show how this modified Kempe chain algorithm works. I actually haven't even tried to generate examples and I also need to rewrite some parts of the code. These examples graphs will quickly show that this method is false or show that it may be an algorithm that works in practice. As you can see, there's no theorems that support my claims, that's why it's a fun proposition.

Next: generate examples/counterexamples and formal methods.

5.1. **Honours.** Again, this was the culmination of what I was able to understand from W2026 URA with Professor Nehaniv's group. Prof. Nehaniv, Amena, and Hanna helped me learn and taught me many mathematical ideas. They helped me learn Latex as well. The Heawood counterexample was actually 50% formulated for ourselves and I was taught these ideas on minors, degeneracy, etc.! I wasn't able to understand much of Tree Width Theory or graphs&algebra at the end.

6. Wrong Paths Taken

6.1. **My delusion in the Modified Kempe's Method.** I originally thought that the Modified Kempe's Method was very unreliable. It turns out maybe it's not so bad.

This method assumes 2 things:

(1) (some sort of) **Greedy coloring applies**. Since this method incorporates greedy coloring, we need to assume that greedy coloring won't get stuck somewhere. I think this is true, but I'm not sure how to prove it. It's probably quite easy...

(2) "We can cross the bridge". Just like in the case of Greedy coloring. The modified method can get stuck if it can't "cross a bridge", and so it can't color all vertices. Like a modified hamiltonian circuit type of thing where we need to make sure we can touch all the necessary vertices to color the graph. I will explain the method and how the very simple method centres around these two assumptions.

Hyper-Greedy: One can see why this method requires some greedy coloring assumption. We simply just find vertices v_* and then greedily color its neighbours using the same Kempe chain method and assume it won't block. This method is definitely not intuitive, as said by Professor as well. So it may be very faulty, but there is a part of me that thinks it will suffice. I believe that Hadwiger's Conjecture $\iff$ Modified Kempe's Method. I have an argument that supports the greedy coloring. AThen then it kind of works, if it's not then it also works, nly IF we can't cross thebridge. and since we delete vertices, the colored vertices are always on the border so I don't have the bandwidth to investigate further when it fails. I know that knowing when something fails is the biggest indicator of knowledge in a certain field... But we will make this hyper-greedy coloring assumption that it works.

"Crossing the bridge": There are known cases when the modified Kempe method fails. The modified Kempe method fails when the selected v_* results in ***deadlock*** in the graph. But in these known cases, choosing a different v_* can avoid this deadlock. Like a search tree, I assume that there's always a path/branch to a solution. Deadlock looks like the following:

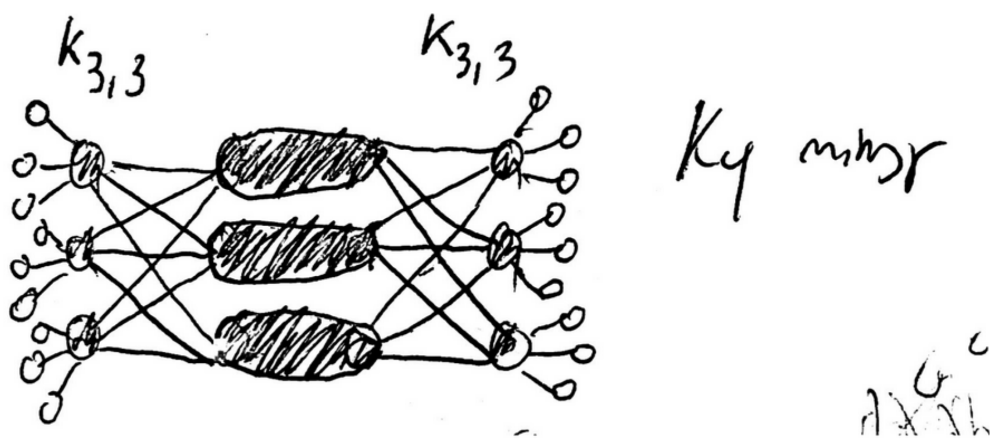


Figure 11. This graph has no K_5 minor. It was formed by joining two $K_{3,3}$s and thus has a K_4 minor. If v_* is chosen such that all $6 * 3 = 18$ "outer" vertices with degree $= 1 \leq 5$ are chosen as v_* first, then the middle 3 vertices will never be colorable (since degree $= 6 > 5$). Therefore the idea in avoiding deadlock is by first choosing an outer vertex, removing it, and then choosing on of the unique $K_{3,3}$ vertices. As you can see, by choosing an alternate path. The bridge is crossed and the entire graph can be colored through streamlinely. No deadlock occurs.

6.2. **Failing Critical's Theorem 4.1.** The counterexample is as follows:

I put my waste of time failure below due to the sunk-cost.

Consider case 2; Since the graph is connected, the minimal degree for any vertex is 1. Assume in G that there are k+1 vertices each with degree 1, and all other vertices have a degree k instead.

$$\implies \sum_{v \in V(G)} \deg(v) = k + (k+1)[v - (k+1)]$$

$$\implies \sum_{v \in V(G)} \deg(v) = k + (k+1)(v - k - 1)$$

$$\implies \sum_{v \in V(G)} \deg(v) = k + (vk - k^2 - k + v - k - 1)$$

$$\implies \sum_{v\in V(G)} \deg(v) = -k^2 + vk + v - k - 1$$

$$\implies \sum_{v\in V(G)} \deg(v) = v(k+1) - k^2 - k - 1$$

By Handshaking Lemma 1.4

$$\implies 2e = v(k+1) - k^2 - k - 1$$

$$\implies e = \frac{v(k+1) - k^2 - k - 1}{2}$$

Therefore, these non null, finite, connected graphs G with no K_t minor have a proposed edge count as follows:

(1) $t = 1, k = 0$ not applicable since the inherent assumption is that there are $k + 1 > 0, k \geq 0$ vertices.
(2) $t = 2, k = 0 \implies e = (v - 1)/2 = 0$ for v = 1.
(3) $t = 3, k = 1 \implies e = (2v - 3)/2 = v - 1.5$.
(4) $t = 4, k = 3 \implies e = (4v - 13)/2 = 2v - 6.5$.
(5) $t = 5, k = 5 \implies e = (6v - 31)/2 = 3v - 15.5$.
(6) $t = 6, k = 7 \implies e = (8v - 57)/2 = 4v - 28.5$.
(7) $t = 7, k = 9 \implies e = (10v - 91)/2 = 5v - 45.5$.

As we can see this degree configuration on G is not possible. Let us take the lower bound as this will cause the highest edge-difference between G and a maximally K_{t-1} graph. Doing so tells us how much more edges we can add. Taking the difference, we obtain:

(1) $t = 1, k = 0$ not applicable since the inherent assumption is that there are $k + 1 > 0, k \geq 0$ vertices.
(2) $t = 2, k = 0 \implies \Delta e = 0$ for v = 1.
(3) $t = 3, k = 1 \implies \Delta e = 1$.
(4) $t = 4, k = 3 \implies \Delta e = 4$.
(5) $t = 5, k = 5 \implies \Delta e = 10$.
(6) $t = 6, k = 7 \implies \Delta e = 19$.
(7) $t = 7, k = 9 \implies \Delta e = 31$. We see that Δe increases by multiples of 3

We proved above that a maximally

(1) $t = 1, k = 0$ not applicable since the inherent assumption is that there are $k + 1 > 0, k \geq 0$ vertices.
(2) $t = 2, k = 0 \implies e = (v - 1)/2 = 0$ for v = 1.
(3) $t = 3, k = 1 \implies e = v - 2$.
(4) $t = 4, k = 3 \implies e = 2v - 5$.
(5) $t = 5, k = 5 \implies e = 3v - 8$.
(6) $t = 6, k = 7 \implies e = 4v - 11$.
(7) $t = 7, k = 9 \implies e = 5v - 14$.

Generalising the trend is as follows. A maximally K_n graph with a k-degeneracy limitation and at least $k + 1$ vertices of degree less than or equal to k only exists if $k \geq n$ or $k \geq t - 1 \implies k + 1 \geq t$

Assume a k-degenerate graph only has 2 vertices (v_1, v_2) with a degree less than or equal to k. Let's assume that these 2 vertices both connect to the same neighbour v_n. v_n would have to have at least $k - 1$ more different neighbours, by looplessness. These $k - 1$ neighbours must also have more than k neighbours each. By finiteness, this is not possible. Now assume that v_1, v_2 don't share a common neighbour. Then the neighbours would have to at least k neighbours each, and again, by finiteness (and looplessness), this is not possible. This trend continues until there are $(k + 1)$ vertices with $\leq$ k vertices. Suppose these $(k + 1)$ vertices each have the same neighbour. This affirms the theorem. Suppose these $(k + 1)$ vertices don't share the same neighbours, then assume that these $(k + 1)$ vertices form K_{k+1}, this affirms the theorem. Suppose that some vertices

7.

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